

GAY SHIN LEE

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ABOUT ME

Adaptable and driven Computer Science student at SUTD with a strong foundation in software development, human-centered design, and artificial intelligence. Experienced in full stack development, and design thinking.

EDUCATION

Singapore University of Technology and Design (SUTD)

- Bachelor of Engineering (Computer Science Design)
- September 2022 - present

CERTIFICATIONS

AWS Certified Cloud Practitioner, Amazon Web Services

Jul 2025

Google UX Design Professional Certificate, Coursera

Sep 2024

TECHNICAL SKILLS

Programming languages : Python, Java, SQL, JavaScript

Operating Systems: Windows, Linux

Development Tools: VSCode, PyCharm, IntelliJ, Jupyter, Git

UI/UX Tools: Figma

Languages: Fluent in English and Chinese language (Written & Spoken)

WORK EXPERIENCES

HTX (Home Team Science & Technology Agency)

Singapore

UX Researcher (Lead)

Sept 2025 - Present

- Led a user-centric discovery with SPF frontline officers (N=10) using contextual inquiry and semi-structured interviews, and synthesised finding via Affinity Mapping and MoSCoW.
- Translated research into user persona, MVP scope, and iterative low-fidelity wireframes.

The National Kidney Foundation (NKF)

Singapore

UX Researcher & UI/UX Designer (Lead)

Sept 2025 - Present

- Led a mixed research (N=61) and translated solution concepts to stakeholders via Pugh Matrix and MoSCoW.
- Planned and authored a moderated usability study for an NKF WebAR mini-game, aligning objectives to stakeholders, ISO 9241-11 and NN/g heuristics.

TechCreate Group

Singapore

Full Stack Software Engineer Intern

May 2025 - Present

- Real Time Engine (FAST Instant Payments) – Singapore Finance Company
 - Delivered a custom RTE for FAST instant payments (ISO 20022), achieving a consistent 10-minute turnaround per file and saving 300+ man-minutes per month.
 - Built a file-format conversion pipeline for ISO 20022 messages, standardizing test data generation and reducing manual prep errors.
 - Designed and executed Postman test suites across 20+ critical payment APIs to raise test coverage and reliability.
- Soundbox Device Activation – Cambodia Bank
 - Authored a comprehensive Functional Specification Document (FSD), translating business requirements into system behaviors, payment flows, error handling, and ISO mappings.
 - Productionized a Python activation script that automates merchant authentication, QR issuance, and deterministic config packaging—cutting per-device activation time from ~2 hours to <15 minutes (~88% reduction).
 - Reduced onboarding variability and increased operational throughput for device activations.
- Regional Rewards Systems – Singapore Bank
 - Mapped end-to-end cross-regional user journeys and presented them to align product, engineering, and business stakeholders across regions.
 - Clarified feature scope and dependencies to accelerate delivery and reduce rework during rollout.

DBS

Singapore

Quality Assurance Intern

Oct 2024 - Jan 2025

- QA testing for DRA (Dynamic Risk Assessment) team using Selenium Cucumber framework with Java for automation test scripts, and SQL for manual testing, which helped reduce test code redundancy.
- Provided mock data based on rule logic and QA descriptions using Pandas.

PROJECTS

50.006 User Interface Design and Implementation - FlowTutor

Sept 2025 - Present

- Applied mixed method research and synthesised 2 user personas, 3 scenario archetypes (Easy/Hard/Edge)
- Translated insights into lo-fi paper prototypes, and hi-fi Figma prototype

Self-initiated

UX Design Project – Reimagining CaringBridge

Sep 2024

- Applied UX research, empathy mapping, user journey analysis, wireframing, and prototyping to deliver a more inclusive and accessible experience.
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CO-CURRICULAR ACTIVITIES

SUTD

Singapore

Chamber Ensemble

Sept 2022 - Dec 2024

Vice President

- Worked with my team to organise for multiple events including SUTD Open House, Lunchtime performances, Mini Showcase etc, and handled various admin matters including the club's finances.

TMJC

Singapore

Photography Club

Jan 2019 - Oct 2021

Art Director

- Designed and planned the layout of art galleries, along with the selection and printing of pictures.
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ADDITIONAL INFORMATION

SUTD FACT Programme (DES x UGM)

Yogyakarta, Indonesia

Jan 2024

- Collaborated with the student in Gadjah Mada University (UGM) to plan and install a solar panel in Turban Village for the village to have power to pump water from the well.

Yonsei International Summer School

Seoul, South Korea

Korean Language

July 2023

- Acquired a foundational understanding of Korean grammar and vocabulary.